UH Hilo Stories

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UH Hilo marine scientists recommend wastewater treatment plant for Puakō

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The researchers have concluded that without a wastewater treatment plant, 69 percent of the Puakō area they sampled would be out of compliance with health department water quality standards.

UH Hilo students, from left, Carmen Garson-Shumway, Saria Adan Sultan, and Christopher Thompson collect nearshore water quality samples to document sewage pollution along the Puakō shoreline in the summer of 2016. This data was part of a recently published study concluding that without a wastewater treatment plant, 69 percent of the area sampled would be out of compliance with state health department water quality standards even if every resident upgrades to new sewage systems. (Photo: Tracy Wiegner/Department of Marine Science/UH Hilo)

By Jordan Hemmerly/UH Hilo Stories.

A dedicated team of researchers at the University of Hawaiʻi at Hilo studying the impact of local sewage systems on coastal water quality has helped confirm Puakō community's need for a wastewater treatment plant. Puakō is located on the west side of Hawaiʻi Island.

Collecting data onsite for several years, the researchers found that on the Puakō coastline, bacterial levels were higher than Hawaiʻi Department of Health standards in recreational waters fronting 81 percent of residential homes sampled.

Steve Colbert (Photo: Raiatea Arcuri/UH Hilo Stories)

Tracy Wiegner (Courtesy photo)

The research team, led by UH Hilo marine science professors **Tracy Wiegner** and **Steve Colbert**, also found that the bacterial levels were elevated regardless of the type of onsite sewage disposal system, cesspool, septic

tank, or aerobic treatment unit the home used.

“We really aren’t the most fun people to go to the beach with,” quips Colbert. “[But] the community has been so supportive of our work, helping us access sites and participating in community meetings where we shared our results.”

The researchers have concluded that without a wastewater treatment plant, 69 percent of the area they sampled would be out of compliance with health department water quality standards even if every resident upgrades to new sewage systems. “From a water quality standpoint, both in regards to human and coral reef health, Puakō building its own [sewage treatment plant] would be best,” state the researchers in their published work.

The study is published in the *Journal of Hydrology: Regional Studies* in an article titled “Identifying locations of sewage pollution within a Hawaiian watershed for coastal water quality management actions” (Dec. 2021).

Working with the community

The research team’s involvement with the Puakō community followed the 2015 ban of construction of new cesspools due to concerns about threats to human and coral reef health. In 2017, state laws were changed to require all cesspools be replaced by 2050.

“Puakō Community Association expressed to us that they were looking to transition away from cesspools,” says Wiegner. “We wanted to help understand the reality of each of these options.”

“We went north, south, and up the mountain to three different communities to figure out where sewage was entering into the groundwater in Puako’s watershed.”

Wiegner and Colbert say the community is considering three options: Replacing cesspools with aerobic treatment units, building a sewage line to the Mauna Lani Resort wastewater treatment plant, or building a treatment center in Puakō.

Challenging location

Puakō is particularly vulnerable to changes in sea level because there is not much distance from the surface of the ground to where water sits below it. This shoreline has one of the highest bacterial concentrations on Hawai’i Island.

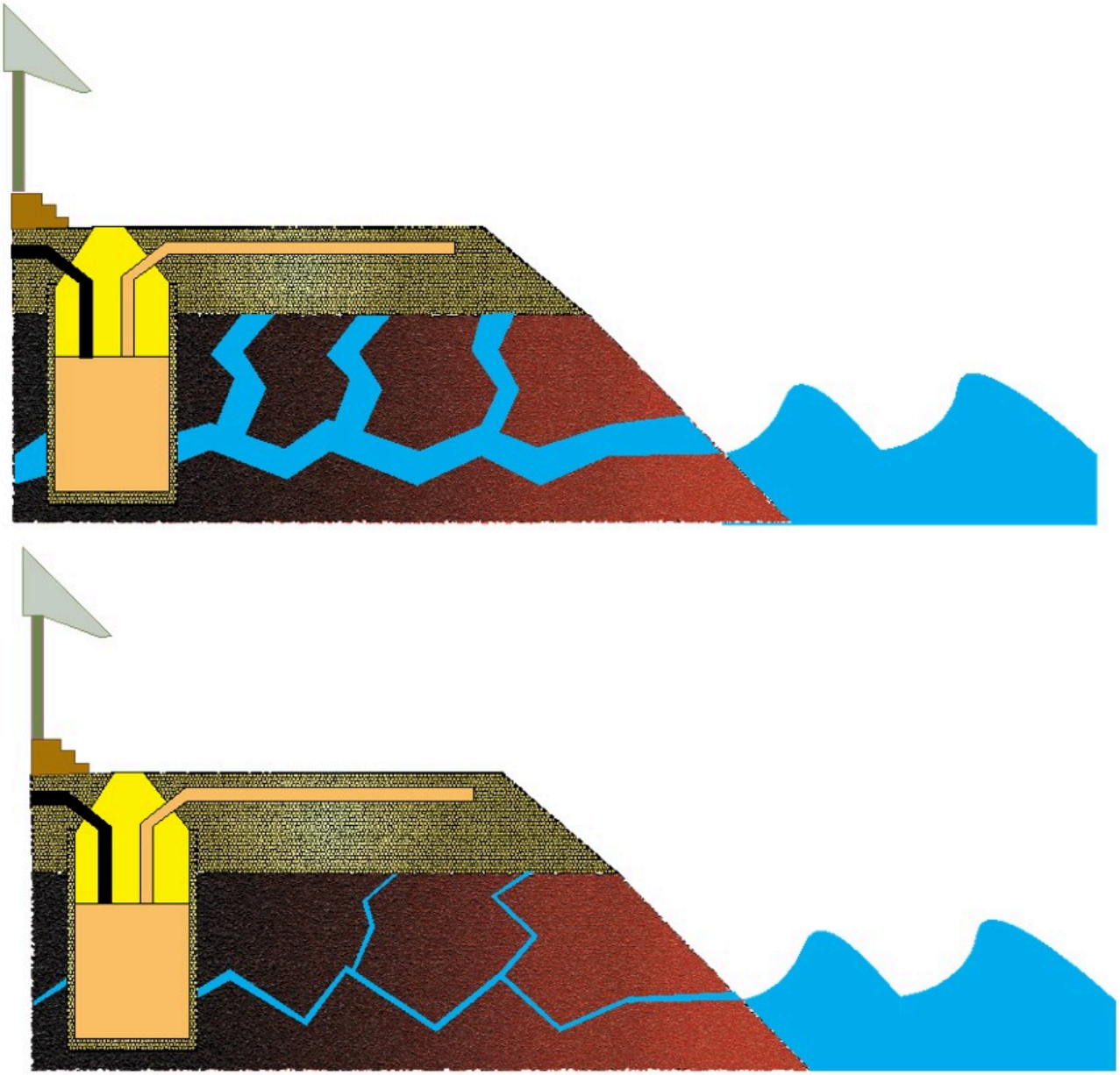
“You’ve got a main road, and houses on either side of it,” Wiegner explains. “There are homes just feet away from the ocean and near tide pools.”

Fluorescein dye emerges from shoreline springs along the Puakō shoreline following a dye tracer test at residential home. Dye tracer studies were conducted in the summer of 2016 to determine the hydrological connection of cesspools, septic tanks, and aerobic treatment units to nearshore waters. (Photo: Steve Colbert/Department of Marine Science/UH Hilo)

The magnitude of sewage pollution increases in areas where there is limited soil, which aids in bacterial processing.

Dye tracer studies were conducted to determine the hydrological connection of cesspools, septic tanks, and aerobic treatment units to nearshore waters.

“In some places, cesspools and septic tanks were placed in areas with large or abundant fractures in the basalt that allow the sewage to flow to the shoreline in a matter of hours,” Colbert says.



The travel time of sewage from onsite sewage disposal systems in Puakō is largely determined by the underlying geology of the property rather than the type of cesspool, septic tank, or aerobic treatment unit at the home. Sewage can reach the shoreline quickly when the basalt has large fractures that connect and make a straight shot, or fairly straight shot, to the shoreline, whereas when the fractures are smaller, it takes a longer time to reach the nearshore waters. Regardless, sewage from the homes is reaching the nearshore waters. (Illustration by Steve Colbert)

Known consequences

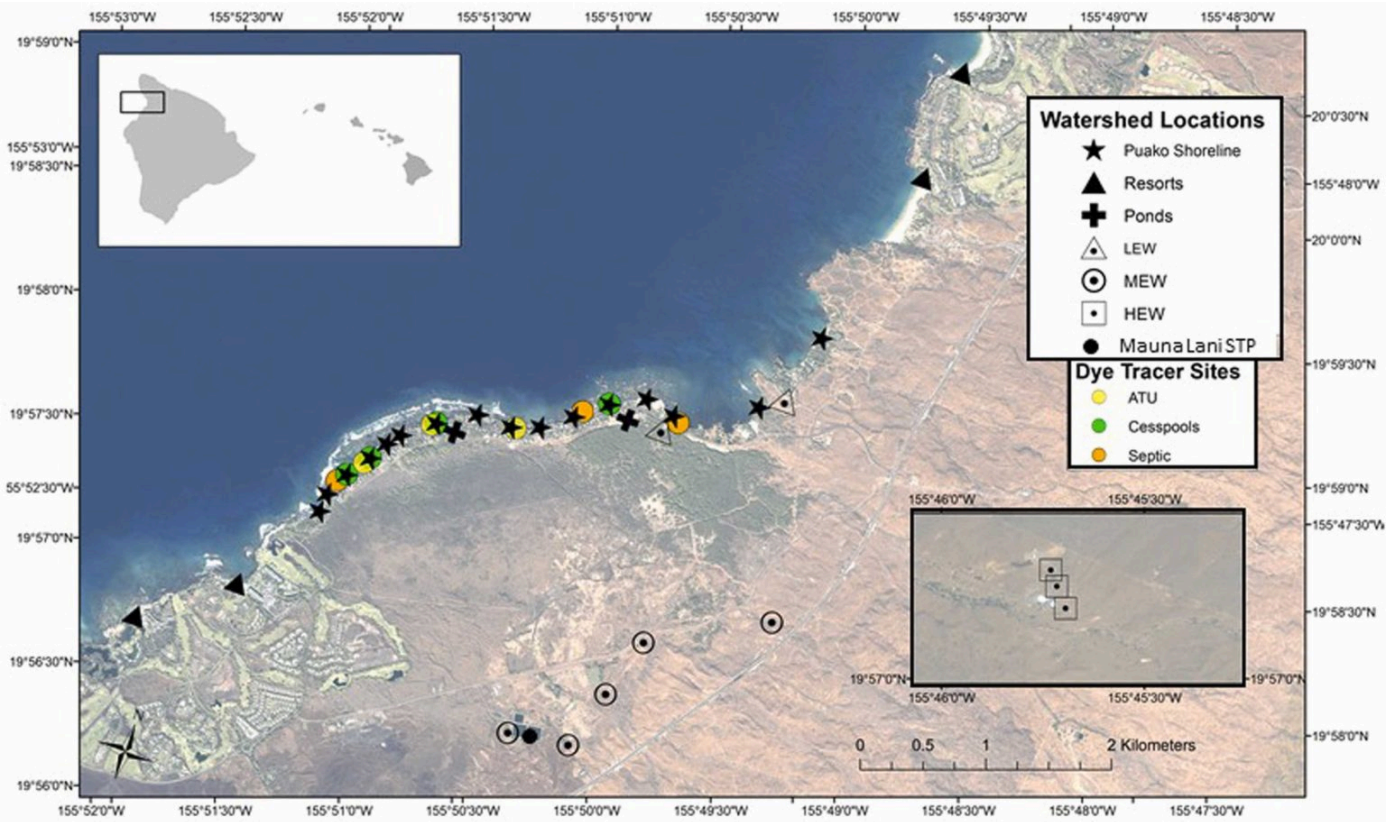
Sewage is composed of a slurry of potentially hazardous pathogens, nutrients, cleaning chemicals, hydrocarbons, and pharmaceuticals. It poses human health risks that can lead to abdominal, skin, urinary, and blood infections.

The state health department publicly reports unsafe bacteria levels in water quality advisory reports posted on their Clean Water Branch System website.

Consistently elevated concentrations of sewage on reefs can stimulate bioerosion, decline of reef diversity, high disease prevalence and severity, shifts in species distributions, and loss of coral community diversity.

Providing data to lawmakers

Water quality data was collected by sampling waters from groundwater wells at Puakō, Waikoloa Village, and Mauna Lani, and from resorts' shorelines at Mauna Kea, Hapuna Prince (now Westin Hapuna Beach Resort), Fairmont Orchid, and Mauna Lani.



The researchers set up 16 stations (stars) along the Puakō coastline, Hawai'i Island, and two resort stations (black triangles) north and south of Puakō. The locations were chosen to evaluate fecal bacteria, nutrients, and nitrate levels in algae. In addition, groundwater samples were collected at high (open squares), mid (open circles), and low (open triangles) elevations and analyzed for sewage indicators. Dye tracer studies were conducted at 10 different locations (yellow circles). (Image: Department of Marine Science/UH Hilo)

Wiegner says she was happy to provide information to lawmakers relevant to the community's requests for improved pollution control in their nearshore waters.

In 2019, the Hawai‘i State Legislature allocated $1.5 million in capital improvement project funds for the planning and design of a wastewater treatment plant at Puakō.

The County of Hawai‘i provided $250,000 in matching funds and encumbered the state’s contributions in order for the planning and design of a wastewater treatment plant in Puakō to move forward.

Dedicated teamwork

Wiegner credits the positive impact of the study to teamwork.

Working alongside Wiegner and Colbert were UH Hilo alumnae **Leilani Abaya** and **Jazmine Panelo**, UH Mānoa Associate Researcher **Craig Nelson**, and Mānoa graduate student **Kristina Remple**.

- Related story: Tracy Wiegner, marine scientist teaches students and communities about coastal water quality (*UH Hilo Stories*, Feb. 11, 2020)

There were numerous supportive projects related to this pursuit, each conducted as a large-scale team effort that included many UH Hilo alumni who have published their findings in peer reviewed journals and as master of science theses.

- Related story: UH Hilo scientists document how rainfall brings harmful bacteria into Hilo Bay (*UH Hilo Stories*, Oct. 23, 2019)

In addition to the Puako Community Association, other partners are the Coral Reef Alliance, The Nature Conservancy, South Kohala Coastal Partnership, Fairmont Orchid Development, Mauna Kea Resort, County of Hawai‘i Department of Water Supply, and UH Mānoa.

“Collaborative efforts like ours provided insight on the sources and impacts of sewage pollution from cesspools in the area,” says Weigner.

B*y Jordan Hemmerly, who is earning her bachelor’s degree in marine science at UH Hilo.*

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